

Progetto n.2

In questo progetto ci sono state assegnate due “task” da completare: la prima era quella di prendere familiarità con i Linux, shell e prompt comandi; mentre la seconda era quella di eseguire un brute-force su un servizio SSH in linguaggio C/Python.

File e directory

In questo primo esercizio dovevamo completare un mini-game intitolato **Gameshell**. Questo conteneva al suo interno delle “missioni” che servono a prendere familiarità con i vari comandi della shell. All’interno del gioco troviamo due comandi che ci permettono di vedere l’obiettivo della missione e il completamento della missione: **gsh goal** (obiettivo); **gsh check** (completamento missione).

Missione 1.

La prima missione è molto semplice: bisognava spostarsi dalla cartella in cui iniziamo alla cartella “Top of the tower”. Inizialmente usiamo **pwd** per capire in quale cartella ci troviamo, subito dopo, con un alternarsi di **ls** (serve per vedere il contenuto di una cartella, che sia una sottocartella o un file) e **cd** (serve per spostarsi tra le cartelle. Aggiungendo il percorso dopo cd ci spostiamo in quella specifica cartella altrimenti torniamo alla cartella home).

```
[use 'gsh help' to get a list of available commands]
[mission 1] $ pwd
/home/kali/gameshell/World

[use 'gsh help' to get a list of available commands]
[mission 1] $ ls
Castle Forest Garden Mountain Stall

[use 'gsh help' to get a list of available commands]
[mission 1] $ cd castle
bash: cd: castle: No such file or directory

[use 'gsh help' to get a list of available commands]
[mission 1] $ cd Castle

[use 'gsh help' to get a list of available commands]
[mission 1] $ ls
Cellar Great_hall Main_building Main_tower Observatory

[use 'gsh help' to get a list of available commands]
[mission 1] $ cd Main_Tower
bash: cd: Main_Tower: No such file or directory

[use 'gsh help' to get a list of available commands]
[mission 1] $ cd Main_tower

[use 'gsh help' to get a list of available commands]
[mission 1] $ ls
First_floor

[use 'gsh help' to get a list of available commands]
[mission 1] $ cd First_floor

[use 'gsh help' to get a list of available commands]
[mission 1] $ ls
Second_floor

[use 'gsh help' to get a list of available commands]
[mission 1] $ cd Second_floor

[use 'gsh help' to get a list of available commands]
[mission 1] $ ls
Top_of_the_tower

[use 'gsh help' to get a list of available commands]
[mission 1] $ cd Top_of_the_tower

[use 'gsh help' to get a list of available commands]
[mission 1] $ gsh check

Congratulations, mission 1 has been successfully completed!
```

Missione 2.

In questa seconda missione l'obiettivo era quello di spostarsi dalla cartella "Top of the tower" alla cartella "Cellar". Per tornare indietro di una cartella utilizziamo, come possiamo vedere dall'immagine, il comando **cd ..** e per spostarsi nelle varie directory utilizziamo di nuovo uno scambio tra i comandi **ls** e **cd** più nome directory.

Missione 3.

```
[use 'gsh help' to get a list of available commands]
[mission 3] $ cd

[use 'gsh help' to get a list of available commands]
[mission 3] $ cd

[use 'gsh help' to get a list of available commands]
[mission 3] $ cd

[use 'gsh help' to get a list of available commands]
[mission 3] $ cd

[use 'gsh help' to get a list of available commands]
[mission 3] $ cd /home/kali/gameshell/World/Castle/Main_building/Throne_room

[use 'gsh help' to get a list of available commands]
[mission 3] $ gsh check

Congratulations, mission 3 has been successfully completed!
```

In questa missione dobbiamo tornare alla cartella di partenza e da lì con massimo due comandi dobbiamo arrivare alla cartella "Throne room". In foto: **cd** per tornare alla cartella iniziale e **cd** più il percorso per arrivare direttamente alla cartella assegnata.

Missione 4.

```

Mission goal
=====
Build a "Hut" in the forest, and then build a "Chest" in the hut.

Useful commands
=====

mkdir DIRECTORY
Create a new directory inside the current directory.
Remark: "mkdir" is an abbreviation for "make directory".

~
[mission 4] $ ls
Castle Forest Garden Mountain Stall

~
[mission 4] $ cd Forest

~/Forest
[mission 4] $ mkdir Hut

~/Forest
[mission 4] $ cd Hut

~/Forest/Hut
[mission 4] $ mkdir Chest

~/Forest/Hut
[mission 4] $ gsh check

Congratulations, mission 4 has been successfully completed!
```

L'obiettivo in questa missione era quello di creare due directory: "Hut" dentro "Forest" e "Chest" dentro "Hut". Ci spostiamo dentro la directory "Forest" e da qui usiamo il comando **mkdir** (serve per creare una directory) per creare "Hut". Subito dopo ci spostiamo dentro "Hut" e sempre con il comando **mkdir** creiamo "Chest".

Missione 5.

```

~/Forest/Hut
[mission 5] $ cd

~
[mission 5] $ ls
Castle Forest Garden Mountain Stall

~
[mission 5] $ cd Castle

~/Castle
[mission 5] $ ls
Cellar Great_hall Main_building Main_tower Observatory

~/Castle
[mission 5] $ cd Cellar

~/Castle/Cellar
[mission 5] $ ls
barrel_of_apples bat_1 bat_2 spider_1 spider_2 spider_3

~/Castle/Cellar
[mission 5] $ rm spider_1 spider_2 spider_3

~/Castle/Cellar
[mission 5] $ ls
barrel_of_apples bat_1 bat_2

~/Castle/Cellar
[mission 5] $ gsh check

Congratulations, mission 5 has been successfully completed!
```

Qui come obiettivo bisognava entrare nella directory "Cellar" ed eliminare i file contenenti il nome "spider". Perciò ci spostiamo nella directory "Cellar" sempre tramite **cd** più percorso e una volta dentro usiamo il comando **ls**. Il comando ci mostra diversi file tra cui quelli con il nome "spider", successivamente usiamo **rm** più il nome del file che vogliamo rimuovere, **rm** infatti sta per "remove".

Missione 6.

```
~/Forest/Hut
[mission 6] $ ls
Chest

~/Forest/Hut
[mission 6] $ cd /home/kali/gameshell/World/Garden

~/Garden
[mission 6] $ ls
coin_1 coin_2 coin_3 Flower_garden Maze Shed

~/Garden
[mission 6] $ mv coin_1 coin_2 coin_3 /home/kali/gameshell/World/Forest/Hut/Chest

~/Garden
[mission 6] $ cd /home/kali/gameshell/World/Forest/Hut/Chest

~/Forest/Hut/Chest
[mission 6] $ ls
coin_1 coin_2 coin_3

~/Forest/Hut/Chest
[mission 6] $ gsh check

Congratulations, mission 6 has been successfully completed!
```

In questa missione aveva come obiettivo di trovare i coin dentro la directory “Garden” e spostare il file coin nella directory “Chest”. Perciò, una volta dentro la cartella “Garden” usiamo il comando **ls** per controllare quanti coin ci sono e dopo con il comando **move** spostiamo i coin nella directory “Chest”.

Missione 7.

```
~/Forest/Hut/Chest
[mission 7] $ cd /home/kali/gameshell/World/Garden

~/Garden
[mission 7] $ ls -A
.11153_coin_2 .17543_coin_3 .48449_coin_1 Flower_garden Maze Shed

~/Garden
[mission 7] $ ./
Flower_garden/ Maze/ Shed/

~/Garden
[mission 7] $ mv .11153_coin_2 .17543_coin_3 .48449_coin_1 /home/kali/gameshell/World/Forest/Hut/Chest

~/Garden
[mission 7] $ cd /home/kali/gameshell/World/Forest/Hut/Chest

~/Forest/Hut/Chest
[mission 7] $ ls -A
.11153_coin_2 .17543_coin_3 .48449_coin_1 coin_1 coin_2 coin_3

~/Forest/Hut/Chest
[mission 7] $ gsh check

Congratulations, mission 7 has been successfully completed!
```

Qui, come nella missione precedente, bisogna spostare i file coin nella directory “Chest”. Unica differenza che troviamo è l’utilizzo del comando **ls -A** perché qui i file erano nascosti.

Missione 8.

```
removed '18035_spider_5'
removed '18502_spider_30'
removed '18814_spider_48'
removed '20331_spider_46'
removed '21482_spider_20'
removed '22576_spider_44'
removed '2348_spider_45'
removed '23864_spider_41'
removed '24222_spider_39'
removed '24232_spider_29'
removed '24452_spider_4'
removed '2476_spider_36'
removed '25919_spider_47'
removed '28082_spider_16'
removed '28126_spider_40'
removed '2862_spider_37'
removed '29297_spider_24'
removed '29953_spider_13'
removed '30348_spider_6'
removed '30601_spider_26'
removed '30670_spider_34'
removed '31441_spider_23'
removed '32692_spider_10'
removed '343_spider_42'
removed '3922_spider_15'
removed '7646_spider_9'
removed '7737_spider_7'
removed '79_spider_25'
removed '8246_spider_14'
removed '9538_spider_32'
removed '9963_spider_3'

~/Castle/Cellar
[mission 8] $ ls -A
23360_bat_5 27240_bat_2 29078_bat_3 31121_bat_4 6757_bat_1 barrel_of_apples

~/Castle/Cellar
[mission 8] $ gsh check
```

In questa missione siamo tornati nella directory “Cellar” per eliminare altri “spider” file. Qui abbiamo nuovamente usato il comando **rm** con l’aggiunta di **-v** in modo da vedere ciò che il comando ha eseguito.

Missione 9.

```
~/Castle/Cellar
[mission 9] $ ls -l
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .10161_spider_42
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .1241_hat_4
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .14503_spider_16
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .17450_spider_4
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .19300_spider_17
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .23565_spider_36
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .24717_spider_38
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .27256_spider_8
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .29078_hat_3
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .30694_spider_40
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .3239_spider_47
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .6846_spider_21
-rw-r--r-- 1 root root 4096 Jan  1 12:00 barrel_of_apples
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .10552_spider_50
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .12924_spider_20
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .14655_spider_26
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .18106_spider_7
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .20050_spider_27
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .23572_spider_25
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .26449_spider_13
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .27645_hat_1
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .29110_spider_10
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .31121_hat_4
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .4870_spider_33
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .6992_spider_29
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .1199_spider_15
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .13354_spider_31
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .14988_spider_19
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .18259_spider_44
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .21735_spider_49
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .23828_spider_28
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .26679_hat_5
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .27686_spider_39
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .29786_spider_46
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .31921_spider_23
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .5367_spider_24
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .7235_spider_9
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .1231_spider_6
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .1372_spider_30
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .16387_spider_5
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .1861_spider_14
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .21819_hat_2
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .24233_spider_43
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .26835_spider_41
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .27963_spider_12
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .29842_spider_18
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .31983_hat_3
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .6055_spider_3
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .774_spider_1
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .12362_spider_35
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .1396_spider_48
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .17139_spider_32
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .19285_spider_34
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .23360_hat_5
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .24252_spider_22
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .27240_hat_2
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .2833_spider_2
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .30210_spider_37
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .3202_spider_45
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .6757_hat_1
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .778_spider_11

~/Castle/Cellar
[mission 9] $ rm .wspiderv

~/Castle/Cellar
[mission 9] $ ls -l
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .1241_hat_4
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .21819_hat_2
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .23360_hat_5
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .26679_hat_5
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .27240_hat_2
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .27645_hat_1
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .29078_hat_3
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .31121_hat_4
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .31983_hat_3
-rw-r--r-- 1 root root 4096 Jan  1 12:00 .6757_hat_1
-rw-r--r-- 1 root root 4096 Jan  1 12:00 barrel_of_apples

~/Castle/Cellar
[mission 9] $ gsh check

Congratulations, mission 9 has been successfully completed!
```

Missione identica a quella precedente solo che qui i file erano nascosti quindi abbiamo usato **ls -A**.

Missione 10.

```
~/Castle/Cellar
[mission 10] $ cd

~
[mission 10] $ cd Castle/Great_hall

~/Castle/Great_hall
[mission 10] $ ls
11392_suit_of_armour 45826_decorative_shield 8026_stag_head standard_1 standard_2 standard_3 standard_4

~/Castle/Great_hall
[mission 10] $ cp standard_1 standard_2 standard_3 standard_4 /home/kali/gameshell/World/Forest/Hut/Chest

~/Castle/Great_hall
[mission 10] $ cd

~
[mission 10] $ cd Forest/Hut/Chest

~/Forest/Hut/Chest
[mission 10] $ ls
coin_1 coin_2 coin_3 standard_1 standard_2 standard_3 standard_4

~/Forest/Hut/Chest
[mission 10] $ gsh check

Congratulations, mission 10 has been successfully completed!
```

In questa missione bisognava copiare i file “standard” nella directory “Chest”. Una volta dentro “Great hall” usiamo **ls** per controllare i file al suo interno, dopodiché usiamo **cp** nome file e percorso dove vogliamo copiare i file.

Missione 11.

```
[mission 11] $ cd Castle/Great_hall
~/Castle/Great_hall
[mission 11] $ ls -A
13159_decorative_shield  31800_tapestry_09      45599_tapestry_05      53472_tapestry_01      7117_tapestry_06      standard_3
20169_tapestry_07        34354_suit_of_armour   47021_tapestry_10      57316_tapestry_08      standard_1             standard_4
30119_tapestry_03        36899_tapestry_02      52855_stag_head        62433_tapestry_04      standard_2

~/Castle/Great_hall
[mission 11] $ cp *tapestry* ~/Forest/Hut/Chest

~/Castle/Great_hall
[mission 11] $ ls ~/Forest/Hut/Chest
20169_tapestry_07  36899_tapestry_02  53472_tapestry_01  7117_tapestry_06  coin_3  standard_3
30119_tapestry_03  45599_tapestry_05  57316_tapestry_08  coin_1            standard_1  standard_4
31800_tapestry_09  47021_tapestry_10  62433_tapestry_04  coin_2            standard_2

~/Castle/Great_hall
[mission 11] $ gsh checl
Error: unknown gsh command 'checl'.
Use one of the following commands:  check, goal, help, reset

~/Castle/Great_hall
[mission 11] $ gsh check

Congratulations, mission 11 has been successfully completed!
```

Anche qui abbiamo dovuto copiare dei file da “Great hall” a “Chest”. Come avvenuto precedentemente, l’unica differenza sta nell’utilizzo del comando **ls -A**.

Missione 12.

```
~/Castle/Main_tower/First_floor
[mission 12] $ ls
painting_CKUswwpQ  painting_WToisKCI  painting_xevhEtBl  Second_floor/

~/Castle/Main_tower/First_floor
[mission 12] $ ls -l
total 16
-rw-rw-r-- 1 kali kali 1055 May 12 2002 painting_CKUswwpQ
-rw-rw-r-- 1 kali kali 1503 Mar 14 2019 painting_WToisKCI
-rw-rw-r-- 1 kali kali 1455 Feb 23 1985 painting_xevhEtBl
drwxrwxr-x 3 kali kali 4096 Nov 17 09:56 Second_floor/

~/Castle/Main_tower/First_floor
[mission 12] $ cp painting_xevhEtBl /Forest/Hut/Chest
cp: cannot create regular file '/Forest/Hut/Chest': No such file or directory

~/Castle/Main_tower/First_floor
[mission 12] $ cp painting_xevhEtBl ~/Forest/Hut/Chest

~/Castle/Main_tower/First_floor
[mission 12] $ cat painting_CKUswwpQ
180da707be132bc21bb46de22538a61b95ae2320@56556
{
  "id": "180da707be132bc21bb46de22538a61b95ae2320",
  "name": "The Great Hall",
  "description": "The Great Hall is a large, ornate room with a high ceiling and a large fireplace. It is the main hall of the castle and is used for formal events and gatherings. The room is decorated with tapestries and paintings, and has a large chandelier hanging from the ceiling. The fireplace is made of dark stone and has a large mantelpiece. The room is very grand and impressive."
}
```

Anche in questa missione ci viene chiesto di copiare dei file nella cartella “Chest”. Qui però utilizziamo due nuovi comandi: **ls -l** che ci permette di vedere i file nel dettaglio, vedendo anche la data dell’ultima modifica; **cat** che ci permette di vedere il contenuto di un file.

Missione 13.

```
~/Castle/Main_tower/First_floor
[mission 13] $ cal 1968

      1968
January February March
Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa
 1  2  3  4  5  6      1  2  3      3  4  5  6  7  8  9
 7  8  9 10 11 12 13    4  5  6  7  8  9 10    10 11 12 13 14 15 16
14 15 16 17 18 19 20    11 12 13 14 15 16 17    17 18 19 20 21 22 23
21 22 23 24 25 26 27    18 19 20 21 22 23 24    24 25 26 27 28 29 30
28 29 30 31            25 26 27 28 29          31

April May June
Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa
 1  2  3  4  5  6      1  2  3  4      1  2  3  4  5  6  7
 7  8  9 10 11 12 13    5  6  7  8  9 10 11    2  3  4  5  6  7  8
14 15 16 17 18 19 20    12 13 14 15 16 17 18    9 10 11 12 13 14 15
21 22 23 24 25 26 27    19 20 21 22 23 24 25    16 17 18 19 20 21 22
28 29 30              26 27 28 29 30 31          23 24 25 26 27 28 29
                                          30

July August September
Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa
 1  2  3  4  5  6      1  2  3      1  2  3  4  5  6  7
 7  8  9 10 11 12 13    4  5  6  7  8  9 10    8  9 10 11 12 13 14
14 15 16 17 18 19 20    11 12 13 14 15 16 17    15 16 17 18 19 20 21
21 22 23 24 25 26 27    18 19 20 21 22 23 24    22 23 24 25 26 27 28
28 29 30 31            25 26 27 28 29 30 31    29 30

October November December
Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa
 1  2  3  4  5      1  2      1  2  3  4  5  6  7
 6  7  8  9 10 11 12    3  4  5  6  7  8  9    8  9 10 11 12 13 14
13 14 15 16 17 18 19    10 11 12 13 14 15 16    15 16 17 18 19 20 21
20 21 22 23 24 25 26    17 18 19 20 21 22 23    22 23 24 25 26 27 28
27 28 29 30 31          24 25 26 27 28 29 30    29 30 31

~/Castle/Main_tower/First_floor
[mission 13] $ gsh check
What was the day of the week for the 11-18-1968?
 1 : Monday
 2 : Tuesday
 3 : Wednesday
 4 : Thursday
 5 : Friday
 6 : Saturday
 7 : Sunday
Your answer: 1
Congratulations, mission 13 has been successfully completed!
```

Nella missione 13 dovevamo capire quale giorno della settimana era l'11/18/1968. Per poterlo fare basta usare il comando **cal** e l'anno che si vuole controllare. Da esso uscirà il calendario di tutto quell'anno e possiamo capire il giorno richiesto dalla consegna.

Missione 14.

```
~
[mission 14] $ ls -A
.bashrc  Castle/  Forest/  Garden/  .lessht  Mountain/  Stall/

~
[mission 14] $ alias la = 'ls -A'
bash: alias: la: not found
bash: alias: =: not found
bash: alias: ls -A: not found

~
[mission 14] $ alias la='ls -A'

~
[mission 14] $ la
.bashrc  Castle/  Forest/  Garden/  .lessht  Mountain/  Stall/

~
[mission 14] $ gsh check

Congratulations, mission 14 has been successfully completed!
```

In questa missione dobbiamo abbreviare il comando **ls -A** in modo da renderlo più semplice e rapido. Per fare ciò usiamo **alias**, la parola semplificata e cosa va a semplificare.

Missione 15.

```
~  
[mission 15] $ cd Forest/Hut/Chest  
  
~/Forest/Hut/Chest  
[mission 15] $ nano journal.txt  
  
~/Forest/Hut/Chest  
[mission 15] $ gsh check  
  
Congratulations, mission 15 has been successfully completed!
```

Nella missione 15 dovevamo creare nella cartella “Chest” un file intitolato “journal.txt”. Per crearlo usiamo **nano** e il nome del file che vogliamo creare.

Missione 16.

```
~  
[mission 16] $ alias journal='nano ~/Forest/Hut/Chest/journal.txt'  
  
~  
[mission 16] $ journal  
  
~  
[mission 16] $ gsh check  
  
Congratulations, mission 16 has been successfully completed!
```

Anche qui, per semplificare il percorso per aprire file “journal.txt” usiamo un **alias**. Quindi digitiamo **alias**, la parola semplificata e cosa va a semplificare.

Missione 17.

```
~/Castle/Cellar  
[mission 17] $ cd .Lair_of_the_spider_queen\ dSHUVEFMxyQNXxCT URKDBMQivFtCnCnV/  
  
~/Castle/Cellar/.Lair_of_the_spider_queen dSHUVEFMxyQNXxCT URKDBMQivFtCnCnV  
[mission 17] $ la  
kWfzVdHmSqPMjsqt_baby_bat_fSYaxAFuIQysCGXI SubdjSRKtDIEvBeM_spider_queen_UGakgXjODJgnRySf  
  
~/Castle/Cellar/.Lair_of_the_spider_queen dSHUVEFMxyQNXxCT URKDBMQivFtCnCnV  
[mission 17] $ rm -r SubdjSRKtDIEvBeM_spider_queen_UGakgXjODJgnRySf  
  
~/Castle/Cellar/.Lair_of_the_spider_queen dSHUVEFMxyQNXxCT URKDBMQivFtCnCnV  
[mission 17] $ gsh check  
Perfect, it took you only 14 seconds to complete this mission!  
  
Congratulations, mission 17 has been successfully completed!
```

Questa era una prova a tempo: dovevamo entrare nella directory nascosta, individuata grazie all’ **alias** **la**, “Lair of the spider queen” e rimuovere definitivamente tramite comando **rm -r** il file “spider queen”.

Missione 18.

```
~  
[mission 18] $ xeyes  
^C  
~  
[mission 18] $ xeyes  
^C  
~  
[mission 18] $ xeyes &  
[1] 48976  
~  
[mission 18] $ ^C  
~  
[mission 18] $ gsh check  
  
Congratulations, mission 18 has been successfully completed
```

In questa missione bisogna semplicemente usare il comando **xeyes** per far apparire un paio di occhi sullo schermo. Per interrompere il loro processo bastava digitare il comando **ctrl c**.

Missione 19.

[illegible]

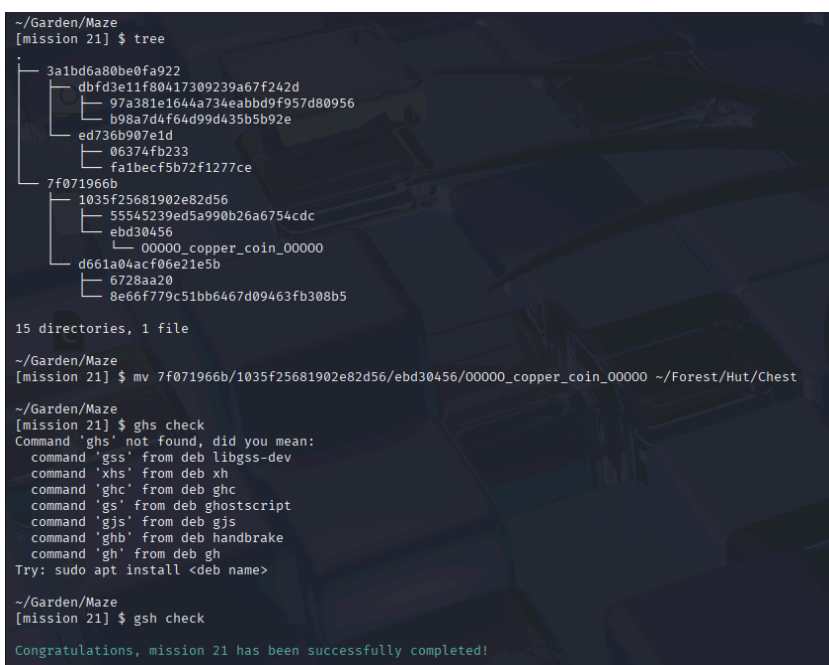
In questa missione ci viene richiesto di far vedere uno spettacolo di fuochi d'artificio. Per far scoppiare più fuochi contemporaneamente usiamo **&** in modo da avviare più programmi nello stesso periodo. Quindi usiamo il comando **gsh check** in modo che il nostro spettatore possa vedere i fuochi d'artificio e successivamente avviamo, tramite comando **flamingo**, lo spettacolo.

Missione 20.



In questa missione dobbiamo aiutare il pirotecnico del re a ricordare la formula magica per il gran finale del suo spettacolo. Per farlo dobbiamo usare il comando **charmiglio** più 4 lettere casuali. Dopo qualche tentativo fallito al quarto tentativo ci siamo riusciti e l'incantesimo è andato a buon fine!

Missione 21.



Nella missione 21 dobbiamo trovare una moneta di rame nel “labirinto” e spostarlo su “Chest”. Innanzitutto ci spostiamo nella cartella “Maze” e da qui usiamo il comando **tree** per vedere tutte le cartelle e sottocartelle fino a trovare “copper coin”. Una volta trovato usiamo **mv** il percorso intero dove si trova copper coin e poi il percorso dove lo vogliamo spostare.

Missione 22.

```
~/Garden/Maze
[mission 22] $ tree
├── 227165780cb5b8a63195569b8
│   ├── 3ab40fe56776a9467dadf3fa3ebb55
│   ├── 49d6551aa88ab6a25df7f2d7f0
│   ├── 6d306ca3d
│   └── 7d78a014d560037316e61660f00
├── 44b3380e
│   ├── 72ab47daddb7d9a4b765e015f04e
│   ├── cd11a792f7ab18ec7
│   ├── e4140858124d03b0b87d1a8ae
│   ├── ad6affcfe76d2e9fed707ec2dde
│   ├── 466bb84b7a7a08
│   ├── 52ceb7093382c163e5cd
│   └── 7d32b1b75f2a95e828d176a9bb781b4
├── b31c54c03d063a358bc604f132
│   ├── 70ea1df0131ffa6969647b775d3
│   ├── 2b2411e3
│   ├── 73f7e18a4f06b5ae
│   ├── 9addb04af9919378693407
│   ├── aef26978ff52
│   ├── 41c14c889b9c
│   ├── 985d20866ec260d3deb08a453
│   ├── fe997188f93e
│   ├── b0f7bc9e95
│   ├── 658a52ebaaf1
│   ├── 71eb40952ae72f002147e383ba5035d
│   ├── f1488a30d83892d33c92d
│   └── 00000_silver_coin_00000
├── ccb70f1c2c0b1480e3a3a4f0687ce
│   ├── af465499317695690ee53d35eada
│   ├── 4e948cd4
│   ├── 803ae1e3d7389c1e7f
│   ├── 8744e8701dcca01ca
│   ├── cca0d92e6d7e330
│   ├── 0a4e120c
│   ├── c87f35be2bde7d7dbfa167577209f2
│   ├── cc112f0932a69450a73a
│   ├── efb667f2cb0a0b9f53d111
│   ├── 12c004cd335e31fbd0d41eb4514983a
│   ├── ac64dc9e900f4344c3fbff2b161be4f
│   └── b7e40d2707
└── 40 directories, 1 file

~/Garden/Maze
[mission 22] $ mv b31c54c03d063a358bc604f132/b0f7bc9e95/f1488a30d83892d33c92d/00000_silver_coin_00000 ~/Forest/Hut/Chest

~/Garden/Maze
[mission 22] $ gsh check
```

Identica missione alla 21 ma con la differenza che invece di un “copper coin” file dobbiamo muovere un “silver coin” file nella cartella “Chest”.

Missione 23.

```
~/Garden/Maze
[mission 23] $ find -iname '*oro*'

~/Garden/Maze
[mission 23] $ find -iname '*gold*'
./4b7ae961ccf8147eb1fa4/2d71e520d32dd/ca109b0ff0dadb83b6adb2e82/GoLD_CoiN_2
./22fe0ee70dd09362f2e8606d97/af9be15c3de3d/841b2b7a68ff4e37237242d9a302/gold_coin_1

~/Garden/Maze
[mission 23] $ mv ./4b7ae961ccf8147eb1fa4/2d71e520d32dd/ca109b0ff0dadb83b6adb2e82/GoLD_CoiN_2 ~/Forest/Hut/Chest

~/Garden/Maze
[mission 23] $ mv ./22fe0ee70dd09362f2e8606d97/af9be15c3de3d/841b2b7a68ff4e37237242d9a302/gold_coin_1 ~/Forest/Hut/Chest

~/Garden/Maze
[mission 23] $ gsh check

Congratulations, mission 23 has been successfully completed!
```

Qui, a differenza delle due missioni precedenti, ci viene chiesto di trovare “gold coin” usando il comando **find -iname** più la parola dentro il file e una volta trovato di metterlo nella cartella “Chest”. Usando il comando **find -iname gold** riusciamo a trovare tutti i file che all’interno hanno la parola “gold”, dopodiché li muoviamo nella cartella “Chest”.

Missione 24.

[illegible]

2. Elixir of youth	pages 3-4
3. Philter of love	page 5
4. Bottled death (powerful poison)	page 6
5. Herbal tea	page 7
6. Draft of invisibility	pages 7-8
7. Homeopathic healing potion (part 1)	pages 8-9
8. Homeopathic healing potion (part 2)	page 10
9. Homeopathic healing potion (part 3)	page 11
10. Toadstool stew	page 12
11. Distilled water	page 13
12. King's ale	Page 13

[illegible]

```
~/Mountain/Cave
[mission 24] $ head -n 6 Book_of_potions/page_07
vvvvvvvvvv
Herbal tea
^^^^^^^^^^

1) Boil water.
2) Add herbs from the forest.
3) Let it sit for five minutes and drink while hot.
```

```
~/Mountain/Cave
[mission 24] $ gsh check

Congratulations, mission 24 has been successfully completed!
```

In questa missione ci viene chiesto di spostarci nella cartella “Cave” e di guardare il procedimento per creare “Herbal tea”. Per prima cosa, una volta dentro “Cave”, usiamo **ls** per vedere il contenuto, in seguito usiamo **cat** sul file “table of contents” e possiamo trovare all’interno varie ricette di pozioni. Dopo aver trovato la pozione “Herbal tea” usiamo nuovamente il comando **cat**, oppure il comando **head**, per vedere il contenuto del file (la differenza è che head, a differenza di cat, mostra solo le prime 10 righe del file.)

Missione 25.

```
~/Mountain/Cave
[mission 25] $ cat Book_of_contents
cat: Book_of_contents: No such file or directory

~/Mountain/Cave
[mission 25] $ cat Book_of_potions/table_of_contents
1. Transformation potion _____ pages 1-2
2. Elixir of youth _____ pages 3-4
3. Philter of love _____ page 5
4. Bottled death (powerful poison) _____ page 6
5. Herbal tea _____ page 7
6. Draft of invisibility _____ pages 7-8
7. Homeopathic healing potion (part 1) _____ pages 8-9
8. Homeopathic healing potion (part 2) _____ page 10
9. Homeopathic healing potion (part 3) _____ page 11
10. Toadstool stew _____ page 12
11. Distilled water _____ page 13
12. King's ale _____ page 13

~/Mountain/Cave
[mission 25] $ cat Book_of_potions/page_12
vvvvvvvvvvvvvvvv
Toadstool stew
vvvvvvvvvvvvvvvv

1) Boil water in a cauldron.
2) Add in a few death caps (Amanita phalloides).
3) Also add a few fly agarics (Amanita muscaria).
4) And some destroying angels (Amanita virosa).
5) Mix in a few deadly webcaps (Cortinarius rubellus).
6) Feel free to add in any colourful fungi you have on hand.
7) Let half of the water evaporate.
8) Season with a pinch of salt and a few herbs.
9) Serve hot in a bowl.

~/Mountain/Cave
[mission 25] $ tail -n 9 Book_of_potions/page_12
1) Boil water in a cauldron.
2) Add in a few death caps (Amanita phalloides).
3) Also add a few fly agarics (Amanita muscaria).
4) And some destroying angels (Amanita virosa).
5) Mix in a few deadly webcaps (Cortinarius rubellus).
6) Feel free to add in any colourful fungi you have on hand.
7) Let half of the water evaporate.
8) Season with a pinch of salt and a few herbs.
9) Serve hot in a bowl
```

Missione molto simile a quella precedente solo che questa volta invece del comando **head** abbiamo usato il comando **tail** che ci fa vedere la parte finale del file.

- 2) Add in a few death caps (*Amanita phalloides*).
- 3) Also add a few fly agarics (*Amanita muscaria*).
- 4) And some destroying angels (*Amanita virosa*).
- 5) Mix in a few deadly webcaps (*Cortinarius rubellus*).
- 6) Feel free to add in any colourful fungi you have on hand.
- 7) Let half of the water evaporate.
- 8) Season with a pinch of salt and a few herbs.
- 9) Serve hot in a bowl.

```
~/Mountain/Cave
[mission 25] $ gsh check
```

Missione 26.

```

1) Boil water in a cauldron.
2) Add 3 measures of fluxweed to the cauldron.
3) Add 2 bundles of knotgrass to the cauldron.
4) Stir 4 times, clockwise.
5) Wave your wand then let potion brew for 80 minutes.
6) Add 4 leeches to the cauldron.
7) Crush 2 scoops of lacewing flies to a fine paste.
8) Add 2 measures of the crushed lacewings to the cauldron.
9) Heat for 30 seconds on a low heat.

~/Mountain/Cave
[mission 26] $ cat Book_of_potions/page_01 Book_of_potions/page_02
vvvvvvvvvvvvvvvvvvvvvvvvv
Transformation potion
^^^^^^^^^^^^^^^^^^^^^^^^^^

1) Boil water in a cauldron.
2) Add 3 measures of fluxweed to the cauldron.
3) Add 2 bundles of knotgrass to the cauldron.
4) Stir 4 times, clockwise.
5) Wave your wand then let potion brew for 80 minutes.
6) Add 4 leeches to the cauldron.
7) Crush 2 scoops of lacewing flies to a fine paste.
8) Add 2 measures of the crushed lacewings to the cauldron.
9) Heat for 30 seconds on a low heat.
10) Add 3 measures of boomslang skin to the cauldron.
11) Crush a bicorn horn into a fine powder.
12) Add 1 measure of the crushed horn to the cauldron.
13) Heat for 20 seconds at a high temperature.
14) Wave your wand then let potion brew for 24 hours.
15) Add 1 additional scoop of lacewings to the cauldron.
16) Stir 3 times, counter-clockwise.
17) Split potion into multiple doses, if desired.
18) Add a pieces of the person you wish to become.
19) Wave your wand to complete the potion.

~/Mountain/Cave
[mission 26] $ gsh check

Congratulations, mission 26 has been successfully completed!

```

In questa missione ci viene chiesto di unire due pagine e mostrarle in un unico file di testo. Per fare ciò ci basta mettere un / tra un file ed un altro.

Missione 27.

```
File Actions Edit View Help
7 3) Let the preparation rest overnight.
8 4) The next morning thank and free your little green friend.
9 5) Boil the water and add in a few sticks of oak tree.
10 6) Crush 5 river stones to a fine powder.
11 7) Mix in a third of the powder and stir vigorously.
12 8) Let the preparation rest for a day.
13 9) Add hairs from the tail of a squirrel (willingly given).
14 10) Add the remaining stone powder.
15 11) Stir the potion very vigorously, in all directions.
16 12) Take some time to rest after such an effort.
17 13) Rest a little bit more.
18 14) Even take a nap if you want.
19 15) Add a few larch tree needles for seasoning.
20 16) Drink the potion from the cauldron.
cat: tail: No such file or directory
cat: 16: No such file or directory

~/Mountain/Cave
[mission 27] $ cat Book_of_potions/page_03 Book_of_potions/page_04 | tail -n 16
1) Fill a cauldron with used bath water.
2) Put a moderately large frog in the water.
3) Let the preparation rest overnight.
4) The next morning thank and free your little green friend.
5) Boil the water and add in a few sticks of oak tree.
6) Crush 5 river stones to a fine powder.
7) Mix in a third of the powder and stir vigorously.
8) Let the preparation rest for a day.
9) Add hairs from the tail of a squirrel (willingly given).
10) Add the remaining stone powder.
11) Stir the potion very vigorously, in all directions.
12) Take some time to rest after such an effort.
13) Rest a little bit more.
14) Even take a nap if you want.
15) Add a few larch tree needles for seasoning.
16) Drink the potion from the cauldron.

~/Mountain/Cave
[mission 27] $ gsh check

Congratulations, mission 27 has been successfully completed!
```

La missione 27 è molto simile alla missione 26. Qui, a parte essere cambiato il numero delle pagine da inserire, ci viene chiesto di usare il comando **tail -n**, il quale ci permette di visualizzare tot numero di righe finale, in questo caso abbiamo preso le ultime 16 righe.

Missione 28.

```
File Actions Edit View Help
7. Homeopathic healing potion (part 1) _____ pages 8-9
8. Homeopathic healing potion (part 2) _____ page 10
9. Homeopathic healing potion (part 3) _____ page 11
10. Toadstool stew _____ page 12
11. Distilled water _____ page 13
12. King's ale _____ Page 13

~/Mountain/Cave
[mission 28] $ cat Book_of_potions/page_13
vvvvvvvvvvvvvvv
Distilled water
^^^^^^^^^^^^^^^^
1) Boil water in a big pot.
2) Condense the vapor in a fresh container.
3) Add minerals for a better taste (optional).
vvvvvvvvvv
King's ale
^^^^^^^^^^^^^^^^
1) Go to the castle's cellar and grab a bottle.
2) Serve cold, in a pint.

~/Mountain/Cave
[mission 28] $ head -n 6 Book_of_potions/page_13
vvvvvvvvvvvvvvv
Distilled water
^^^^^^^^^^^^^^^^
1) Boil water in a big pot.
2) Condense the vapor in a fresh container.
3) Add minerals for a better taste (optional).

~/Mountain/Cave
[mission 28] $ head -n 6 Book_of_potions/page_13 | tail -n 3
1) Boil water in a big pot.
2) Condense the vapor in a fresh container.
3) Add minerals for a better taste (optional).

~/Mountain/Cave
[mission 28] $ gsh check

Congratulations, mission 28 has been successfully completed!
```

In questa missione dobbiamo controllare la ricetta per l'acqua distillata, la quale si trova nella pagina 13 del "Book of potions". Oltre alla ricetta per l'acqua distillata, se controlliamo il contenuto della pagine tramite comando **cat**, ci viene data anche la ricetta per la birra del Re. Visto che a noi interessa solo quella dell'acqua distillata useremo il comando **head** e poi la pagina che ci serve.

Missione 30.

```
ps
  PID TTY          TIME CMD
 1553 pts/0    00:00:00 zsh
 1757 pts/0    00:00:00 bash
 1835 pts/0    00:00:00 bash
 15278 pts/0   00:00:00 spell
 15921 pts/0   00:00:00 spell
 19071 pts/0   00:00:00 spell
 20059 pts/0   00:00:00 ps

~/Mountain/Cave
[mission 30] $
*#*#*
B_***/~
!$-#

*#*#*
B_***/~
!$-#

kill -s
*#*#*
B_***/~
!$-#

9
*#*#*
B_***/~
!$-#

*#*#*
B_***/~
!$-#

*#*#*
B_***/~
!$-#

15
*#*#*
B_***/~
!$-#

278

~/Mountain/Cave
[mission 30] $
```

In questa missione ci viene chiesto di fermare un "sortilegio". Per fermarlo usiamo innanzitutto il comando **ps** in modo da vedere tutti i processi in esecuzione e successivamente usiamo **kill** seguito dal numero del processo per arrestare il processo.

Missione 31.

```
~/Castle/Cellar
[mission 31] $ ps

  PID TTY          TIME CMD
 1553 pts/0    00:00:00 zsh
 1757 pts/0    00:00:00 bash
 1835 pts/0    00:00:00 bash
 21114 pts/0    00:00:00 nice_fairy
 21116 pts/0    00:00:00 mischievous_imp
 21121 pts/0    00:00:00 spell
 21122 pts/0    00:00:00 spell
 21123 pts/0    00:00:00 spell
 21124 pts/0    00:00:00 tail
 21138 pts/0    00:00:00 spell
 21143 pts/0    00:00:00 spell
 21144 pts/0    00:00:00 spell
 21145 pts/0    00:00:00 tail
 52435 pts/0    00:00:00 sleep
 52454 pts/0    00:00:00 sleep
 52521 pts/0    00:00:00 sleep
 52526 pts/0    00:00:00 sleep
 52600 pts/0    00:00:00 sleep
 52614 pts/0    00:00:00 sleep
 52615 pts/0    00:00:00 ps

~/Castle/Cellar
[mission 31] $ pstree 21116
mischievous_imp─3*[spell──sleep]
                  └─tail

~/Castle/Cellar
[mission 31] $ pstree -p 21116
mischievous_imp(21116)─spell(21138)──sleep(63144)
                        └─spell(21143)──sleep(62983)
                        └─spell(21144)──sleep(63036)
                        └─tail(21145)

~/Castle/Cellar
[mission 31] $ kill 21138 21143 21144

~/Castle/Cellar
[mission 31] $ pstree -p 21116
mischievous_imp(21116)─tail(21145)
```

```
~/Mountain/Cave
[mission 31] $ cd

[mission 31] $ cd Castle/Cellar

~/Castle/Cellar
[mission 31] $ ls
10430_coal      15014_coal      20633_snowflake 24443_snowflake 28517_snowflake 33599_snowflake 38462_coal      42885_coal      46077_snowflake 51392_coal      56206_coal      61829_snowflake 65315_snowflake
10443_snowflake 15156_coal      20773_snowflake 24448_snowflake 28564_coal      33689_snowflake 38833_coal      42907_coal      46087_coal      5139_coal      56587_coal      61921_snowflake 6536_coal
1047_coal      15156_snowflake 20818_coal      24449_coal      28564_snowflake 33689_snowflake 39022_coal      42902_snowflake 46087_snowflake 51464_snowflake 56526_coal      61927_coal      65452_coal
10574_snowflake 1516_snowflake 20874_snowflake 2455_coal      28582_snowflake 33689_snowflake 39042_snowflake 42993_snowflake 46321_coal      51479_coal      56578_snowflake 61944_snowflake 65486_coal
10621_snowflake 15253_coal      20875_coal      24568_snowflake 28766_coal      33711_snowflake 39060_snowflake 43023_coal      46480_snowflake 51479_snowflake 56616_coal      62029_coal      6665_snowflake
10798_snowflake 1524_snowflake 2087_snowflake 24591_coal      28844_snowflake 33759_snowflake 39075_snowflake 43200_coal      46514_snowflake 51482_snowflake 56619_coal      62245_snowflake 6669_coal
10831_coal      15434_snowflake 20971_snowflake 24721_coal      28859_snowflake 33797_snowflake 39097_snowflake 43260_snowflake 46550_snowflake 51480_coal      56638_coal      62300_snowflake 6757_bat_1
10848_coal      15502_coal      21113_coal      24802_coal      28899_coal      33960_snowflake 3910_coal      43358_coal      46559_snowflake 51654_snowflake 56658_coal      62370_snowflake 6769_coal
1093_coal      15502_snowflake 21143_coal      24802_snowflake 28930_snowflake 33973_coal      39125_coal      43446_coal      46705_coal      51775_coal      56876_coal      62476_snowflake 6944_snowflake
10962_snowflake 15529_snowflake 21321_snowflake 24843_coal      28930_snowflake 33996_coal      39168_coal      43450_snowflake 4699_snowflake 51824_snowflake 56947_coal      62555_coal      7104_snowflake
11039_coal      15555_snowflake 21524_coal      24877_coal      29086_snowflake 34008_coal      39386_snowflake 43522_coal      47023_snowflake 51881_coal      56987_coal      62686_coal      7132_coal
1110_snowflake 15888_snowflake 21687_coal      24929_coal      29105_coal      33973_coal      39168_coal      43522_coal      47023_snowflake 51889_coal      56987_coal      62686_coal      7132_coal
11192_coal      1589_snowflake 21874_snowflake 24974_snowflake 29122_coal      34122_coal      39570_snowflake 43723_snowflake 47165_coal      51889_snowflake 57149_snowflake 62687_coal      7211_snowflake
11232_coal      15949_coal      21719_snowflake 2501_snowflake 29078_bat_3 34161_snowflake 39753_snowflake 43746_snowflake 47240_coal      51927_snowflake 57286_coal      62709_coal      7245_snowflake
11284_snowflake 16319_snowflake 21759_coal      2511_coal      29317_coal      34180_coal      39936_snowflake 43818_snowflake 47397_snowflake 51972_coal      57286_snowflake 62724_coal      7309_snowflake
11305_coal      16358_snowflake 21884_coal      2519_coal      29349_snowflake 34349_snowflake 4005_snowflake 43883_coal      47522_coal      52196_snowflake 57292_coal      62802_coal      7322_snowflake
11367_coal      16456_coal      21926_coal      25471_snowflake 29394_coal      3450_snowflake 40150_coal      43912_coal      47548_coal      521_snowflake      57298_snowflake 62894_snowflake 7381_coal
11676_coal      16460_snowflake 21982_coal      25522_snowflake 29394_snowflake 34627_coal      4026_coal      44020_coal      47606_snowflake 52259_snowflake 57310_snowflake 62901_coal      73_coal
11676_coal      16629_coal      21982_coal      25529_snowflake 29398_coal      347_coal      40462_coal      44020_snowflake 52330_snowflake 57351_coal      62921_snowflake 7314_snowflake
11707_coal      16722_coal      22033_snowflake 25556_snowflake 29705_coal      34846_coal      40499_coal      44078_coal      47731_snowflake 52377_coal      57559_coal      62935_coal      7528_snowflake
11788_coal      16970_coal      22090_snowflake 25568_coal      29726_coal      35248_coal      4057_snowflake      44118_coal      47924_snowflake 52505_snowflake 57597_snowflake 63115_coal      7659_snowflake
11745_snowflake 16990_coal      22192_coal      2566_coal      2014_snowflake 35509_coal      40721_snowflake 40593_coal      47065_snowflake 52684_coal      57633_snowflake 63127_coal      7164_coal
11773_coal      17065_snowflake 22290_coal      25719_coal      30365_coal      35525_snowflake 40807_snowflake 441_snowflake      48057_snowflake 52661_snowflake 58237_coal      63103_coal      776_snowflake
11766_coal      17243_coal      22365_snowflake 25789_coal      30426_snowflake 35545_snowflake 40833_coal      44245_snowflake 48147_snowflake 52677_coal      58254_snowflake 63109_coal      7970_coal
11804_snowflake 17616_coal      22502_coal      25937_coal      30451_coal      35577_coal      40941_coal      44332_snowflake 48180_snowflake 52683_coal      58412_coal      63201_snowflake 805_coal
11828_snowflake 17717_snowflake 22765_snowflake 26089_snowflake 30466_snowflake 35640_coal      40955_coal      44388_snowflake 48199_coal      52697_snowflake 58452_snowflake 6340_coal      805_snowflake
11940_snowflake 1773_coal      22774_snowflake 26058_snowflake 30546_coal      35746_snowflake 40961_snowflake 44388_snowflake 4841_coal      52838_coal      58463_coal      63419_snowflake 8099_coal
12045_coal      17777_coal      22840_coal      2611_snowflake 3065_coal      36033_snowflake 41002_snowflake 44688_coal      48439_coal      52876_coal      58519_coal      63441_coal      8111_snowflake
12076_coal      1778_coal      2288_coal      26280_coal      3066_snowflake 36126_coal      41095_coal      44730_snowflake 48452_coal      52884_snowflake 58629_coal      6348_coal      8344_snowflake
1208_snowflake 17845_coal      22935_snowflake 26280_snowflake 30757_snowflake 36216_coal      41108_snowflake 44838_coal      48480_coal      52911_snowflake 5863_coal      6348_snowflake 8739_snowflake
12226_snowflake 17945_coal      22940_coal      26294_coal      30873_snowflake 36216_snowflake 41412_coal      44960_snowflake 48571_coal      53059_snowflake 58771_snowflake 6386_snowflake 8877_snowflake
1254_snowflake 18076_coal      22964_coal      26308_snowflake 30894_coal      36358_snowflake 41508_snowflake 44960_snowflake 48587_coal      53059_snowflake 58829_coal      6388_snowflake 8892_snowflake
12656_snowflake 18130_snowflake 23067_coal      26464_coal      31805_coal      36358_snowflake 41508_snowflake 44960_snowflake 48587_coal      53059_snowflake 58829_coal      6388_snowflake 8892_snowflake
12677_coal      18296_coal      23076_snowflake 26526_snowflake 31811_coal      36406_snowflake 41505_snowflake 44960_snowflake 48587_coal      53059_snowflake 58829_coal      6388_snowflake 8892_snowflake
12691_coal      18500_snowflake 23260_coal      26572_coal      31821_bat_4 36454_snowflake 41628_coal      41515_coal      44960_snowflake 48587_coal      53059_snowflake 58829_coal      6388_snowflake 8892_snowflake
1305_coal      1858_snowflake 23306_snowflake 26554_snowflake 31844_snowflake 36672_snowflake 41647_snowflake 45318_snowflake 4888_snowflake 53582_snowflake 58829_coal      6388_snowflake 8892_snowflake
13124_coal      18686_coal      23313_coal      26599_snowflake 31839_coal      36700_coal      41770_snowflake 45411_coal      48932_snowflake 53591_coal      5890_coal      64096_snowflake 9278_coal
13249_snowflake 18679_coal      23322_coal      26655_snowflake 31897_snowflake 36874_snowflake 4184_coal      45424_coal      49190_snowflake 53712_snowflake 58925_snowflake 64306_snowflake 9341_snowflake
13250_coal      18672_snowflake 23322_snowflake 27104_coal      31514_snowflake 37126_coal      41912_snowflake 45439_coal      49424_snowflake 53901_coal      58946_coal      64308_snowflake 9366_coal
13250_snowflake 18741_snowflake 23360_bat_5 27104_snowflake 31604_coal      37129_coal      42026_snowflake 49402_snowflake 53973_snowflake 53973_snowflake 58986_coal      64419_snowflake 9405_coal
13254_coal      18864_snowflake 23491_coal      2711_snowflake 31741_coal      37141_coal      42045_coal      49402_snowflake 53973_snowflake 53973_snowflake 58986_coal      64419_snowflake 9405_coal
13308_snowflake 18871_snowflake 23526_coal      27238_coal      31897_snowflake 37202_snowflake 42139_snowflake 45484_coal      49927_coal      54081_snowflake 59977_snowflake 64426_coal      9727_coal
13348_snowflake 18959_coal      23528_coal      27238_snowflake 31914_coal      37221_snowflake 42140_snowflake 45484_snowflake 50022_snowflake 54122_coal      60002_coal      64426_snowflake 9766_snowflake
13500_coal      18987_coal      23576_snowflake 27249_bat_2 31969_snowflake 37284_snowflake 42140_snowflake 45520_snowflake 50156_coal      54190_snowflake 60012_snowflake 64432_snowflake 9770_snowflake
13722_coal      19093_coal      23703_coal      27261_coal      3201_coal      37285_snowflake 4219_snowflake 4557_coal      50249_coal      54470_coal      60012_snowflake 64449_snowflake 9787_coal
13755_coal      19114_coal      23780_snowflake 27261_snowflake 32080_coal      37337_snowflake 42200_snowflake 45673_snowflake 50425_coal      54478_snowflake 60045_coal      64460_coal      9793_coal
```

Nella missione 31 bisogna trovare dentro la cartella “Cellar” ed eliminare il file “mischievous imp” e “coal”. Usando il comando **ps** troviamo tutti i processi in esecuzione e da qui possiamo trovare il codice del processo “mischievous imp”. Una volta trovato il codice lo inseriamo dopo il comando **pstree** in modo da avere un sistema ad albero delle cartelle e sottocartelle di “mischievous imp”, se aggiungiamo un **-p** prima del codice riusciamo ad avere i codici di tutte le sottocartelle. Una volta ottenuti usiamo il comando **kill** nelle prime 3 cartelle “spell” (eliminando le prime verranno automaticamente eliminate pure quelle al loro interno). Per rimuovere i file “coil” basta semplicemente usare **rm *coal***, in questo modo tutti i file con all’interno

la parola coal verranno eliminati.

```
~/Castle/Cellar
[mission 31] $ rm coal

~/Castle/Cellar
[mission 31] $ ls
10203_snowflake 14604_snowflake 19474_snowflake 24446_snowflake 289_snowflake 33996_snowflake 39047_snowflake 42400_snowflake 45619_snowflake 49892_snowflake 54190_snowflake 60828_snowflake 64884_snowflake
10453_snowflake 14717_snowflake 19675_snowflake 24448_snowflake 29006_snowflake 34014_snowflake 39042_snowflake 42543_snowflake 45673_snowflake 50022_snowflake 54478_snowflake 60888_snowflake 64_snowflake
10524_snowflake 14832_snowflake 19668_snowflake 24568_snowflake 29078_bat_3 34085_snowflake 39060_snowflake 42551_snowflake 45701_snowflake 50091_snowflake 54710_snowflake 60880_snowflake 65052_snowflake
10621_snowflake 14882_snowflake 19674_snowflake 24802_snowflake 2912_snowflake 34101_snowflake 39075_snowflake 42602_snowflake 45786_snowflake 50102_snowflake 54726_snowflake 60869_snowflake 65105_snowflake
10635_snowflake 14880_snowflake 19681_snowflake 24974_snowflake 29201_snowflake 34219_snowflake 39097_snowflake 42661_snowflake 45855_snowflake 50400_snowflake 54755_snowflake 61399_snowflake 65172_snowflake
10798_snowflake 15060_snowflake 19970_snowflake 2501_ssnowflake 29369_snowflake 34349_snowflake 39117_snowflake 42693_snowflake 45876_snowflake 50419_snowflake 55007_snowflake 61538_snowflake 65315_snowflake
1084_snowflake 15130_snowflake 20091_snowflake 25394_snowflake 29394_snowflake 34449_snowflake 39152_snowflake 42707_snowflake 46085_snowflake 50434_snowflake 55061_snowflake 61829_snowflake 6533_snowflake
10891_snowflake 15156_snowflake 20252_snowflake 25182_snowflake 29764_snowflake 3450_snowflake 39206_snowflake 42902_snowflake 46045_snowflake 50640_snowflake 55197_snowflake 61921_snowflake 6620_snowflake
10962_snowflake 15160_snowflake 20445_snowflake 25348_snowflake 29896_snowflake 34715_snowflake 393_snowflake 42983_snowflake 46077_snowflake 50647_snowflake 55306_snowflake 61944_snowflake 6665_snowflake
11068_snowflake 1516_snowflake 20456_snowflake 25471_snowflake 30013_snowflake 348_snowflake 39367_snowflake 42993_snowflake 46087_snowflake 50706_snowflake 55380_snowflake 62112_snowflake 6757_bat_1
1110_snowflake 1524_snowflake 20623_snowflake 25522_snowflake 30027_snowflake 349_snowflake 39392_snowflake 43116_snowflake 46265_snowflake 50752_snowflake 55472_snowflake 62245_snowflake 6944_snowflake
11241_snowflake 15434_snowflake 20772_snowflake 25629_snowflake 3014_snowflake 35525_snowflake 39396_snowflake 43181_snowflake 46398_snowflake 50861_snowflake 55510_snowflake 62309_snowflake 7094_snowflake
11284_snowflake 15555_snowflake 20874_snowflake 25656_snowflake 30317_snowflake 35545_snowflake 39392_snowflake 43221_snowflake 46480_snowflake 51246_snowflake 55745_snowflake 62370_snowflake 7104_snowflake
11305_snowflake 15584_snowflake 2087_snowflake 23888_snowflake 30426_snowflake 35739_snowflake 39394_snowflake 43269_snowflake 46487_snowflake 51330_snowflake 55844_snowflake 6239_snowflake 7204_snowflake
11313_snowflake 15529_snowflake 20972_snowflake 25920_snowflake 30466_snowflake 35746_snowflake 4005_snowflake 4343_snowflake 46514_snowflake 51347_snowflake 5597_snowflake 62476_snowflake 7208_snowflake
11514_snowflake 15555_snowflake 20977_snowflake 26009_snowflake 30510_snowflake 36833_snowflake 40233_snowflake 43445_snowflake 46550_snowflake 51415_snowflake 5602_snowflake 62620_snowflake 7211_snowflake
11630_snowflake 15647_snowflake 20909_snowflake 26058_snowflake 30529_snowflake 36216_snowflake 40224_snowflake 43453_snowflake 46559_snowflake 51450_snowflake 56098_snowflake 62661_snowflake 7245_snowflake
11715_snowflake 15880_snowflake 21233_snowflake 2611_snowflake 3066_snowflake 36144_snowflake 4057_snowflake 43496_snowflake 46605_snowflake 51464_snowflake 56415_snowflake 62694_snowflake 7283_snowflake
11755_snowflake 15894_snowflake 21691_snowflake 26280_snowflake 30757_snowflake 36358_snowflake 40721_snowflake 43598_snowflake 4699_ssnowflake 51479_snowflake 56570_snowflake 62921_snowflake 7389_snowflake
11804_snowflake 1589_snowflake 21719_snowflake 26526_snowflake 30873_snowflake 36380_snowflake 40731_snowflake 43612_snowflake 47023_snowflake 51482_snowflake 56686_snowflake 63072_snowflake 7372_snowflake
11818_snowflake 16260_snowflake 21743_snowflake 26654_snowflake 31121_bat_4 36440_snowflake 40807_snowflake 43627_snowflake 47085_snowflake 51654_snowflake 56910_snowflake 6314_snowflake 7368_snowflake
11936_snowflake 16319_snowflake 21944_snowflake 26792_snowflake 31144_snowflake 36484_snowflake 4081_snowflake 43713_snowflake 47205_snowflake 51824_snowflake 57098_snowflake 63166_snowflake 7514_snowflake
11940_snowflake 16350_snowflake 22033_snowflake 26899_snowflake 31178_snowflake 36672_snowflake 40831_snowflake 43727_snowflake 47397_snowflake 51889_snowflake 57149_snowflake 63179_snowflake 7528_snowflake
1200_snowflake 16460_snowflake 22040_snowflake 26952_snowflake 31473_snowflake 36795_snowflake 40871_snowflake 43746_snowflake 47606_snowflake 51927_snowflake 57200_snowflake 63201_snowflake 7539_snowflake
12226_snowflake 17000_snowflake 22090_snowflake 27104_snowflake 31489_snowflake 36784_snowflake 40891_snowflake 43818_snowflake 47680_snowflake 51998_snowflake 57255_snowflake 63225_snowflake 7659_snowflake
12300_snowflake 17065_snowflake 22299_snowflake 2711_snowflake 31514_snowflake 36929_snowflake 41002_snowflake 43922_snowflake 47731_snowflake 52100_snowflake 5728_snowflake 63288_snowflake 776_snowflake
12311_snowflake 17225_snowflake 22312_snowflake 27144_snowflake 31713_snowflake 37082_snowflake 41108_snowflake 44016_snowflake 47924_snowflake 52144_snowflake 57290_snowflake 6331_snowflake 805_snowflake
1254_snowflake 1724_snowflake 22365_snowflake 27238_snowflake 31897_snowflake 37221_snowflake 4133_snowflake 44020_snowflake 47965_snowflake 52158_snowflake 57319_snowflake 63311_snowflake 8111_snowflake
12656_snowflake 17389_snowflake 22765_snowflake 27240_bat_2 31969_snowflake 37284_snowflake 41503_snowflake 441_snowflake 48057_snowflake 521_snowflake 57427_snowflake 6348_snowflake 8301_snowflake
12684_snowflake 17712_snowflake 22774_snowflake 27261_snowflake 32049_snowflake 37285_snowflake 41585_snowflake 44245_snowflake 48147_snowflake 52210_snowflake 57597_snowflake 63504_snowflake 8544_snowflake
12805_snowflake 18130_snowflake 22843_snowflake 27806_snowflake 32170_snowflake 37337_snowflake 41607_snowflake 44268_snowflake 48176_snowflake 52259_snowflake 57637_snowflake 63700_snowflake 8739_snowflake
13097_snowflake 18500_snowflake 22935_snowflake 27807_snowflake 32195_snowflake 37348_snowflake 41640_snowflake 4433_snowflake 48180_snowflake 52330_snowflake 57664_snowflake 63806_snowflake 8890_snowflake
13229_snowflake 1858_snowflake 23070_snowflake 27812_snowflake 3227_snowflake 37395_snowflake 41642_snowflake 44300_snowflake 48249_snowflake 52305_snowflake 58254_snowflake 6380_snowflake 8877_snowflake
13250_snowflake 18611_snowflake 23080_snowflake 28190_snowflake 32296_snowflake 37522_snowflake 41647_snowflake 44308_snowflake 48049_snowflake 52361_snowflake 58452_snowflake 63934_snowflake 8922_snowflake
13388_snowflake 18672_snowflake 23322_snowflake 28191_snowflake 32413_snowflake 37530_snowflake 4165_snowflake 446_snowflake 48654_snowflake 52697_snowflake 58704_snowflake 63996_snowflake 8952_snowflake
13440_snowflake 18741_snowflake 23360_bat_5 28222_snowflake 32487_snowflake 37684_snowflake 41770_snowflake 44726_snowflake 48703_snowflake 52789_snowflake 58771_snowflake 64095_snowflake 9140_snowflake
13497_snowflake 18741_snowflake 23520_snowflake 28310_snowflake 32556_snowflake 37910_snowflake 4188_snowflake 44739_snowflake 48871_snowflake 52884_snowflake 5882_snowflake 64096_snowflake 9341_snowflake
13722_snowflake 18817_snowflake 23534_snowflake 28458_snowflake 32752_snowflake 37955_snowflake 41912_snowflake 44783_snowflake 4888_snowflake 52911_snowflake 59028_snowflake 64165_snowflake 9416_snowflake
13729_snowflake 18804_snowflake 23644_snowflake 28494_snowflake 33139_snowflake 38040_snowflake 41925_snowflake 4490_snowflake 48972_snowflake 53019_snowflake 59393_snowflake 6421_snowflake 9479_snowflake
13765_snowflake 18871_snowflake 23760_snowflake 28590_snowflake 33290_snowflake 38069_snowflake 42002_snowflake 44966_snowflake 49008_snowflake 5311_snowflake 59722_snowflake 64305_snowflake 9507_snowflake
13790_snowflake 188_snowflake 23951_snowflake 28517_snowflake 3346_snowflake 38069_snowflake 42139_snowflake 45047_snowflake 49128_snowflake 53337_snowflake 59780_snowflake 64368_snowflake 9683_snowflake
13792_snowflake 18919_snowflake 23977_snowflake 28564_snowflake 33591_snowflake 38118_snowflake 42140_snowflake 45077_snowflake 49190_snowflake 53428_snowflake 59875_snowflake 64419_snowflake 9739_snowflake
13793_snowflake 1900_snowflake 24000_snowflake 28592_snowflake 33599_snowflake 3821_snowflake 4219_snowflake 45147_snowflake 49210_snowflake 5344_snowflake 59977_snowflake 64422_snowflake 9766_snowflake
13819_snowflake 19114_snowflake 24211_snowflake 28702_snowflake 33689_snowflake 38220_snowflake 42200_snowflake 45318_snowflake 49424_snowflake 53582_snowflake 60011_snowflake 64426_snowflake 9770_snowflake
14018_snowflake 19134_snowflake 24332_snowflake 28831_snowflake 3368_snowflake 38288_snowflake 42275_snowflake 45439_snowflake 49462_snowflake 5372_snowflake 60012_snowflake 64432_snowflake 993_snowflake
14062_snowflake 19200_snowflake 24390_snowflake 28844_snowflake 33711_snowflake 38311_snowflake 42290_snowflake 45492_snowflake 49527_snowflake 5375_snowflake 60129_snowflake 64449_snowflake 9900_snowflake
14215_snowflake 19330_snowflake 24422_snowflake 28903_snowflake 33759_snowflake 38578_snowflake 42370_snowflake 45484_snowflake 49718_snowflake 53973_snowflake 60227_snowflake 64594_snowflake barrel_of_appl
14362_snowflake 19472_snowflake 24443_snowflake 28930_snowflake 33968_snowflake 38722_snowflake 42387_snowflake 45528_snowflake 49814_snowflake 54081_snowflake 60230_snowflake 64805_snowflake
```

Missione 32.

```
~/Castle/Cellar
[mission 32] $ gsh check
35 + 62 = ?? 97
45 + 23 = ?? 68
47 + 98 = ?? 145
60 + 61 = ?? 121
93 + 54 = ?? 147

Congratulations, mission 32 has been successfully completed!
```

Questa missione è un semplice quiz di matematica senza limite di tempo. Per iniziarlo bastava semplicemente digitare **gsh check**.

Missione 33.

```
[mission 33] $ cd Castle/Main_building/Library

~/Castle/Main_building/Library
[mission 33] $ gsh check < Mathematics_101
84 * 11 = ?? 6 * 27 = ?? 40 * 95 = ?? 92 * 59 = ?? 87 * 57 = ?? 71 * 87 = ?? 95 * 62 = ?? 22 * 6 = ?? 99 * 100 = ?
? 86 * 100 = ?? 11 * 77 = ?? 3 * 45 = ?? 13 * 62 = ?? 8 * 71 = ?? 94 * 99 = ?? 41 * 83 = ?? 59 * 72 = ?? 72 * 42 =
? 67 * 76 = ?? 22 * 95 = ?? 86 * 29 = ?? 17 * 3 = ?? 32 * 27 = ?? 73 * 64 = ?? 7 * 76 = ?? 74 * 10 = ?? 34 * 47
= ?? 94 * 90 = ?? 52 * 76 = ?? 85 * 53 = ?? 55 * 83 = ?? 95 * 90 = ?? 4 * 96 = ?? 48 * 49 = ?? 3 * 88 = ?? 47 * 82
= ?? 28 * 100 = ?? 1 * 89 = ?? 11 * 80 = ?? 95 * 23 = ?? 54 * 43 = ?? 72 * 89 = ?? 57 * 13 = ?? 56 * 57 = ?? 4 *
91 = ?? 51 * 74 = ?? 76 * 76 = ?? 19 * 30 = ?? 42 * 30 = ?? 78 * 81 = ?? 72 * 60 = ?? 73 * 68 = ?? 51 * 25 = ?? 4
* 46 = ?? 4 * 45 = ?? 22 * 55 = ?? 99 * 87 = ?? 88 * 21 = ?? 4 * 59 = ?? 16 * 26 = ?? 36 * 82 = ?? 33 * 57 = ?? 99
* 67 = ?? 47 * 9 = ?? 57 * 34 = ?? 32 * 47 = ?? 41 * 67 = ?? 11 * 79 = ?? 48 * 18 = ?? 35 * 57 = ?? 86 * 32 = ??
45 * 29 = ?? 71 * 40 = ?? 12 * 85 = ?? 40 * 12 = ?? 98 * 17 = ?? 31 * 94 = ?? 85 * 8 = ?? 86 * 83 = ?? 74 * 99 = ?
? 83 * 1 = ?? 26 * 60 = ?? 11 * 87 = ?? 5 * 81 = ?? 17 * 16 = ?? 21 * 52 = ?? 95 * 37 = ?? 43 * 59 = ?? 31 * 99 =
?? 62 * 30 = ?? 77 * 92 = ?? 94 * 66 = ?? 52 * 4 = ?? 64 * 98 = ?? 82 * 6 = ?? 14 * 5 = ?? 25 * 41 = ?? 57 * 50 =
?? 96 * 66 = ?? 54 * 94 = ??

Congratulations, mission 33 has been successfully completed!
```

Anche questo era un quiz di matematica ma a tempo. Per fare in modo di riuscire a completare il quiz bisogna aggiungere Mathematics 101 dopo il simbolo minore in

modo da usare il contenuto di un file come input per un comando, invece di digitarlo manualmente.

Missione 34.

```
grimoire_11774 grimoire_15244 grimoire_18009 grimoire_21574 grimoire_26216 grimoire_31824 grimoire_8547
grimoire_11999 grimoire_15347 grimoire_18057 grimoire_22130 grimoire_2641 grimoire_31958 grimoire_8863
grimoire_12144 grimoire_15580 grimoire_18232 grimoire_22179 grimoire_26830 grimoire_32466
grimoire_12389 grimoire_15618 grimoire_1849 grimoire_22306 grimoire_27525 grimoire_32734
grimoire_12394 grimoire_16462 grimoire_18699 grimoire_22360 grimoire_2781 grimoire_4352
grimoire_12413 grimoire_16485 grimoire_18965 grimoire_22522 grimoire_2799 grimoire_4404

~/Castle/Main_building/Library/Merlin_s_office
[mission 34] $ ls grimoire* > Drawer/inventory.txt

~/Castle/Main_building/Library/Merlin_s_office
[mission 34] $ less Drawer/inventory.txt

~/Castle/Main_building/Library/Merlin_s_office
[mission 34] $ gsh check

Congratulations, mission 34 has been successfully completed!
```

In questa missione dobbiamo salvare una lista di “spell books” in un file chiamato “inventory.txt”. Come prima cosa inviamo i file “grimoire” dentro un file invece di mostrarlo sullo schermo. Con il comando **less** possiamo vedere il contenuto di un file una pagina per volta.

Missione 35.

```
~/Castle/Main_building/Library/Merlin_s_office
[mission 35] $ grep -l -i gsh grimoire_* 2> /dev/null
grimoire_ANCxfNdHfrP
grimoire_cHxqFSGBoU
grimoire_ESBNpivzrTMcfJBLLVyHKoju
grimoire_eVSFnKjrgoOnJeRzBxx
grimoire_FBFaGuBJmYeP
grimoire_FtFnoPiSEdwxCEOfB
grimoire_hEIgSjKb
grimoire_jGUJtHLuVgYnriGDFVPPeHbZj
grimoire_JoJeOUfk
grimoire_KZKNldlSDZfvCZPb
grimoire_LXMCvtOiBgobmAKFQkx
grimoire_lzsTJZSMdkqydscmcGSvgjanvV
grimoire_mLjwMvhnG
grimoire_OjwKZfQhLfuzURjglYRDtIcQTpwMtfh
grimoire_qyvenrBzvGVROyxQaJPMqWpaMujt
grimoire_rwZznjOBtQn
grimoire_umRinhMRwrpkHqsXaoH
grimoire_uuWNCIjxbJVEwjXAKxZdB
grimoire_vjRPBXPhwRHnAwXImcEpqEBpdmRIEo
grimoire_VZWoxLKThlDTJBQDbNOv
grimoire_WCXBwqkvUXdVLSMKWqAHLZWqTr
grimoire_xFbkqPQf
grimoire_yhxcPsbmedWIqb
grimoire_ZaSFidgQWYWyAhHLKGkztLTR

~/Castle/Main_building/Library/Merlin_s_office
[mission 35] $ gsh check

Congratulations, mission 35 has been successfully completed!
```

In questa missione dobbiamo estrarre una lista di tutti i file che contengono la “composizione alchemica gsh”. I comandi che usiamo qui sono **grep** (serve per cercare una stringa dentro un file e le mette in riga) **-l** (serve per stampare solo i filenames contenente nella stringa) **-i** (non fa differenza tra maiuscola e minuscola) **gsh 2** (manda un messaggio d'errore al file) e **/dev/null** (file virtuale universalmente vuoto).

Missione 37.

```
~/Castle
[mission 37] $ cd Main_building

~/Castle/Main_building
[mission 37] $ ls
Library/ Throne_room/

~/Castle/Main_building
[mission 37] $ cd Throne_room

~/Castle/Main_building/Throne_room
[mission 37] $ ls
Kings_quarter/

~/Castle/Main_building/Throne_room
[mission 37] $ ls -l
total 4
drw-rw-r-- 2 kali kali 4096 Nov 17 09:56 Kings_quarter/

~/Castle/Main_building/Throne_room
[mission 37] $ chmod +x Kings_quarter/

~/Castle/Main_building/Throne_room
[mission 37] $ ls -l
total 4
drwxrwxr-x 2 kali kali 4096 Nov 17 09:56 Kings_quarter/

~/Castle/Main_building/Throne_room
[mission 37] $ cd Kings_quarter

~/Castle/Main_building/Throne_room/Kings_quarter
[mission 37] $ gsh check

Congratulations, mission 37 has been successfully completed!
```

In questa missione dobbiamo riuscire ad avere accesso alla cartella “King’s quarter”. Per fare ciò vediamo innanzitutto i permessi che abbiamo in quella cartella tramite il comando **ls -l**. Per poter accedere dobbiamo renderla "executable" e per farlo usiamo il comando **chmod** che ci permette di modificare i permessi dei vari gruppi. Una volta resa la cartella executable possiamo accedervi.

Missione 38.

```
~/Castle/Main_building/Throne_room/Kings_quarter
[mission 38] $ ls
note

~/Castle/Main_building/Throne_room/Kings_quarter
[mission 38] $ ls -A
note .secret_note

~/Castle/Main_building/Throne_room/Kings_quarter
[mission 38] $ ls -Al
total 8
-rw-rw-r-- 1 kali kali 11 Nov 19 07:26 note
--w--w---- 1 kali kali 11 Nov 19 07:26 .secret_note

~/Castle/Main_building/Throne_room/Kings_quarter
[mission 38] $ chmod +r .secret_note

~/Castle/Main_building/Throne_room/Kings_quarter
[mission 38] $ cat .secret_note
2254147782

~/Castle/Main_building/Throne_room/Kings_quarter
[mission 38] $ gsh check
What's the combination to open the King's safe? 2254147782

Congratulations, mission 38 has been successfully completed!
```

In questa missione bisogna trovare dentro la cartella “King’s quarter” la combinazione per la cassaforte del re. Per trovarlo innanzitutto notiamo che tramite **ls -A** troviamo il file nascosto “.secret_note”, usando **ls -Al** possiamo vedere i permessi del file .secret_note, in questo caso non è possibile leggere il contenuto del file. Per renderlo leggibile usiamo **chmod** e

aggiungiamo la possibilità di leggere con la lettera “r” sul file .secret_note. Una volta fatto ci appare la combinazione richiesta che ci permetterà di andare avanti.

Missione 39.

```
[mission 39] $ cd Safe

~/Castle/Main_building/Throne_room/Safe
[mission 39] $ ls -l
total 4
-- 1 kali kali 48 Nov 19 07:30 crown

~/Castle/Main_building/Throne_room/Safe
[mission 39] $ chmod +r crown

~/Castle/Main_building/Throne_room/Safe
[mission 39] $ mv crown ~/Forest/Hut/Chest
mv: cannot move 'crown' to '/home/kali/gameshell.3/World/Forest/Hut/Chest/crown': Permission denied

~/Castle/Main_building/Throne_room/Safe
[mission 39] $ cat crown
(*^+*^*)
\@*@\@/
{_947_}

~/Castle/Main_building/Throne_room/Safe
[mission 39] $ cd ..

~/Castle/Main_building/Throne_room
[mission 39] $ ls -hl
total 8.0K
drwxrwxr-x 2 kali kali 4.0K Nov 19 07:23 Kings_quarter/
dr-xr-xr-x 2 kali kali 4.0K Nov 19 07:30 Safe/

~/Castle/Main_building/Throne_room
[mission 39] $ chmod +w Safe/

~/Castle/Main_building/Throne_room
[mission 39] $ ls -l
total 8
drwxrwxr-x 2 kali kali 4096 Nov 19 07:23 Kings_quarter/
drwxrwxr-x 2 kali kali 4096 Nov 19 07:30 Safe/

~/Castle/Main_building/Throne_room
[mission 39] $ cd Safe

~/Castle/Main_building/Throne_room/Safe
[mission 39] $ mv crown ~/Forest/Hut/Chest
mv: cannot move 'crown' to '~/Forest/Hut/Chest': No such file or directory

~/Castle/Main_building/Throne_room/Safe
[mission 39] $ mv crown ~/Forest/Hut/Chest/

~/Castle/Main_building/Throne_room/Safe
[mission 39] $ gsh check
What are the 3 digits inscribed on the base of the crown? 947

Congratulations, mission 39 has been successfully completed!
```

Una volta dentro la cassaforte dobbiamo rubare la corona e metterla in “Chest”. Per poterla prendere dobbiamo digitare una sequenza magica di numeri. Per vedere i numeri usiamo il comando **cat** sul file “crown” e possiamo vedere l’immagine di una corona dove alla base ci sono segnate 3 cifre. Per poter spostare la corona dobbiamo cambiare i diritti della cartella (non avendo i diritti per scrivere non ci è possibile spostare la corona) quindi usiamo **chmod** e aggiungiamo “w” alla cartella “Safe”. Infine mettiamo **mv** e il percorso per portarla alla “Chest”.

Missione 40.

```
~/Castle/Main_building/Throne_room/Safe
[mission 40] $

~/Castle/Main_building/Throne_room/Safe
[mission 40] $ cd

~
[mission 40] $ cd Garden

~/Garden
[mission 40] $ cd Maze

~/Garden/Maze
[mission 40] $ ls
003b3d8eb052f92332fe1d39da2a74/  4490d4293d5fd95/  53b9dc8304ada119e3889d30a/  77ddfc4c8bfae48462d8ed2/  e7b2abb679cb0a52f553ef/
22cde79f4e49d9ab3a5cf0/  4a7f99e303506115646/  699ccec03f5e9df/  dc8f517d382ed9d0ca633/  e84320e744770d240b0c87cda/

~/Garden/Maze
[mission 40] $ find . -type f
./699ccec03f5e9df/9a6f8c68558f5cb57e9220/e9259508b4/55896
./699ccec03f5e9df/c0eff22d2568bf/991cd29cf1d9b8140434/31207
./22cde79f4e49d9ab3a5cf0/2f19ce1c62c7f402f2691a9e6/7f8a80631ce1d44a/16229
./22cde79f4e49d9ab3a5cf0/658719a0c8baa59800f7f65d7/9ba4ab03ee66ca93df1d2c250af91/17480
./e7b2abb679cb0a52f553ef/6aa2eae6b8bdf1984d/f52ffba02f3a683bc92de6405332ef0/15999
./4490d4293d5fd95/7c3da132e3b85f33769f6/62fce1e7d6/18937

~/Garden/Maze
[mission 40] $ cat ./699ccec03f5e9df/9a6f8c68558f5cb57e9220/e9259508b4/55896
55896 ruby 4acb2b9992e7821e74a35637943cc9d10406f002

~/Garden/Maze
[mission 40] $ mv ./699ccec03f5e9df/9a6f8c68558f5cb57e9220/e9259508b4/55896 ~/Forest/Hut/Chest/

~/Garden/Maze
[mission 40] $ gsh check

Congratulations, mission 40 has been successfully completed!
```

Nella missione 40 dobbiamo trovare un rubino nel labirinto e spostarlo nella “Chest”. Una volta dentro “Maze” usiamo **ls** per vedere il suo contenuto. Utilizzando il comando **find** possiamo trovare un file o una directory nella directory in cui ci troviamo, aggiungendo **-type f** cerchiamo un file nello specifico

contenente la lettera f al suo interno. Dopo aver trovato i file, guardiamo il loro contenuto usando **cat** e il nome del file fino a quando non troviamo quello che

contiene il rubino. Una volta trovato spostiamo il file con **mv** e lo mettiamo nella cartella “Chest”.

Missione 42

```
(0)
~/Stall
[mission 42] $ grep King *_s_c_r_o_l_l_* | grep -v PAID
the King bought a blanket for 3 coppers.
the King bought a dented helmet for 4 coppers.
the King bought a pick for 4 coppers.
the King bought a bag flour for 2 coppers.
the King bought a wooden spoon for 5 coppers.
the King bought a bag flour for 6 coppers.
(1)
~/Stall
[mission 42] $ gsh check
How much does the king owe? 24

Congratulations, mission 42 has been successfully completed!
```

In questa missione dobbiamo trovare in solo 3 comandi a quanto ammonta il debito del Re. Dentro la cartella “Stall” troviamo molti file con il nome “boring object” perciò dobbiamo filtrarli: usando il comando **ls | grep boring object** ci appariranno tutti i file eccetto quelli con il nome “boring object”. Dopo aver tolto i file “boring object” dobbiamo filtrare anche i file pagati, per farlo usiamo **grep -v PAID**. Una volta filtrati sia i file “boring object” sia i file “PAID” appariranno solo i file del Re non ancora pagati.

Missione 44.

```
~/Castle/Main_building/Library/Merlin_s_office/Drawer
[mission 44] $ cat secret_message | tr v u
user uf zl juvy:
lbh juvy trg zl purfg, naq rirclguat ug pbagnuaf.
guuf purfg uf ua gur pryne, naq gur jbeq gb znxr
ug er-nccrne uf: qnwo
zreyua gur rapunagre

~/Castle/Main_building/Library/Merlin_s_office/Drawer
[mission 44] $ cat secret_message | tr "a-z" "n-za-m"
here is my will:
you will get my chest, and everything it contains.
this chest is in the cellar, and the word to make
it re-appear is: dajb
merlin the enchanter

~/Castle/Main_building/Library/Merlin_s_office/Drawer
[mission 44] $ gsh check
What's the key that will make Merlin's chest to appear?
dajb

Congratulations, mission 44 has been successfully completed!
[ progress was saved in /home/kali/gameshell-save.sh ]
```

Nell’ultima missione dobbiamo decifrare un messaggio segreto. Innanzitutto usiamo il comando **cat** per vedere il contenuto del messaggio segreto. Per sostituire le parole usiamo il comando **tr**: la prima stringa verrà sostituita dalla seconda. Dopo aver “tradotto” il messaggio segreto troviamo la parola segreta che ci permetterà di far apparire la cassa di Merlino concludendo, in questo modo il gioco!



